

Lab Chapter 4

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Part 1: Distribution Functions

```
# 1. Generate a data frame consisting of 10 realizations from a Uniform distribution with lower limit 3  
uniform_realizations <- data.frame(Values = runif(10, min = 3, max = 10))  
print(uniform_realizations)
```

```
##      Values  
## 1  8.593009  
## 2  9.998672  
## 3  6.521784  
## 4  6.342072  
## 5  6.932639  
## 6  3.994007  
## 7  8.205395  
## 8  4.142541  
## 9  7.961587  
## 10 9.373188
```

```
# 2. Find the values of the probability density function for the integers 0 through 3 of an exponential  
dexp(0:3, 2)
```

```
## [1] 2.000000000 0.270670566 0.036631278 0.004957504
```

```
# 3. Find the value that cuts off the bottom 70% of a chi squared distribution with 5 degrees of freedom  
qchisq(0.70, 5)
```

```
## [1] 6.06443
```

```
# 4. Find P(X<=0.6) if X follows a Beta distribution with a=2 and b=4.  
pbeta(0.6, 2, 4)
```

```
## [1] 0.91296
```

```
# 5. Generate a data frame consisting of 20 realizations from a Normal Random Variable with mean 50 and  
normal_realizations <- data.frame(Values = rnorm(20, mean = 50, sd = 10))  
print(normal_realizations)
```

```
##      Values  
## 1  56.58705  
## 2  45.37638  
## 3  46.72489  
## 4  59.39032  
## 5  43.17824  
## 6  35.57542
```

```
## 7 42.51429
## 8 56.05101
## 9 47.54295
## 10 55.46167
## 11 56.06989
## 12 58.87848
## 13 54.41237
## 14 39.63305
## 15 49.61174
## 16 52.48788
## 17 52.22537
## 18 46.09781
## 19 66.99319
## 20 35.81016
```

Part 2: Monte Carlo Approximations

```
set.seed(41)
n <- 100000

# 1. The mean of X+Y if X~Uniform(0,3) and Y~Beta(a=3, b=1) and X and Y are independent
mean(runif(n, 0, 3) + rbeta(n, 3, 1)) #X+Y Mean

## [1] 2.25424
1.5 + 0.75 #Theory

## [1] 2.25
# 2. The variance of W+ 2Z if W~chisq(df=7) Y~Normal(mean=0,sd=2) and W and Z are independent.
var(rchisq(n, 7) + 2 * rnorm(n, 0, 2))#W+2Z Variance: MC

## [1] 30.02572
14 + 16 #Theory

## [1] 30
# 3. P(Q>0.3) if Q is Exponential with rate parameter 5.
mean(rexp(n, 5) > 0.3)#P(Q > 0.3)

## [1] 0.22278
1 - pexp(0.3, 5)#Theory

## [1] 0.2231302
```

Part 3: Complex Monte Carlo Simulations

```
#Approximate the probability that a randomly selected person will score greater than a 90 on an Exam us

study_time <- rexp(n, rate = 1/2) # Exponential
sleep_hours <- runif(n, 4, 9) # Uniform sleep
exam_mean <- 85 + 2 * study_time # Mean score
exam_sd <- sleep_hours / 10 # SD
```

```

exam_scores <- rnorm(n, exam_mean, exam_sd) # Scores
mean(exam_scores > 90)

## [1] 0.29117

#Approximate the probability distribution for the total distance that a bird will migrate(from starting

result <- do(10000)*[
  rain <- rbinom(365, 1, 0.20)#rain
  wind <- rchisq(365, df = 3)#wind

  north_move <- ifelse(rain == 1,rnorm(365, 3, 2),rnorm(365, 10 + wind, 4)) # North dependency
  west_move <- ifelse(rain == 1, rnorm(365, 0, 1),rnorm(365, 1, 3)) # west depenedncy
  sqrt(sum(north_move)^2 + sum(west_move)^2)# Final distance
}

# Create histogram with density
hist(result$result, freq = FALSE, main = "Bird Migration Distance")
mean_dist <- mean(result$result)
sd_dist <- sd(result$result)
print(mean_dist)#mean

## [1] 4028.52

print(sd_dist)#sd

## [1] 111.953

lines(2000:5000, dnorm(2000:5000, mean = mean_dist, sd = sd_dist), col = "red")#normal curve

```

